6CCSACCM: Cooperation, Coordination and Multi-Agent Systems

Week 3

Repeated and extensive-form games

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Normal form representation

- So far we assumed that a game is represented in normal form: effectively, as a big table.
- In some sense, this is reasonable.
  - The normal form is conceptually straightforward.
  - The normal form representation is universal:
    while many other representations exist to describe finite games, each of them has an "induced normal form"—
    a corresponding normal-form representation that preserves game-theoretic properties such as Nash equilibria.

Extensive form games

- **Extensive form** is a finite representation that does not always assume that players act simultaneously.
- This representation is in general exponentially smaller than its induced normal form, and furthermore can be much more natural to reason about.
  - While the Nash equilibria of an extensive form game can be found through its induced normal form, computational benefit can be had by working with the extensive form directly.
  - Furthermore, there are other solution concepts, which explicitly refer to the sequence in which players act and which are therefore not meaningful when applied to normal form games.

Extensive form games

- Extensive form games are widely used in artificial intelligence and multiagent systems to model situations involving sequential decision-making, strategic interaction over time, and the handling of imperfect or incomplete information.
- These representations—often visualized as game trees—are especially important where agents' choices, order of moves, observations, and payoffs unfold dynamically during interaction.

Applications in AI

Sequential Decision-Making:

- Extensive form games serve as the foundation for algorithms used in classic AI problems like computer chess and Go, where the sequential nature of moves and branching possibilities are modeled explicitly.

Planning and Search:

- AI uses game-tree search algorithms such as minimax and alpha-beta pruning derived from extensive form representations to optimise policy selection in adversarial domains.

Applications in AI

Learning and Strategy Optimisation:

- Techniques like extensive-form fictitious self-play and reinforcement learning leverage game trees to learn optimal strategies, especially in environments with imperfect information (e.g., poker).

Automated Negotiation:

- Modelling negotiation protocols as extensive form games helps AI design agents that engage in multi-stage, strategic interaction with partners, including bluffing and information revelation.

Applications in MAS

Multiagent Reinforcement Learning:

- Extensive form games model how multiple agents interact over time, each making decisions based on their private and shared histories. This enables truthful modelling of sequential, stage-dependent strategies and coordination problems.

Resource Allocation and Auctions:

- Sequential auctions, where bids are placed in turn, are naturally described as extensive form games to capture each agent's information and tactical decisions.

Applications in MAS

Security and Defense:

- In patrolling and adversarial scenarios (e.g., robot patrolling, cyber-security), agents decide on sequential actions to maximise system utility, often analysed via extensive form representations.

Imperfect Information Games:

- Many multi-agent robotic or AI systems feature partial observability, modelled as extensive form games with information sets representing agents' uncertainty at each decision point.

Extensive form games with perfect information

- Informally speaking, a perfect-information game in extensive form (or simply, a **perfect-information game**) is a *tree* in the sense of graph theory, in which:
  - each node represents the choice of one of the players,
  - each edge represents a possible action, and
  - the leaves represent final outcomes over which each player has a utility function.

Extensive form games with perfect information

Definition

A (finite) **perfect-information game** (in extensive form) is a tuple $(N, A, H, Z, \chi, \rho, \sigma, u)$, where:

- N is a set of n players;
- A is a (single) set of actions;
- H is a set of nonterminal choice nodes;
- Z is a set of terminal nodes, disjoint from H;

Extensive form games with perfect information

- $\chi : H \to 2^A$ is the action function, which assigns to each choice node a set of possible actions;
- $\rho : H \to N$ is the player function, which assigns to each nonterminal node a player $i \in N$ who chooses an action at that node;
- $\sigma : H \times A \to H \cup Z$ is the successor function, which maps a choice node and an action to a new choice node or terminal node such that for all $h_1, h_2 \in H$ and $a_1, a_2 \in A$, if $\sigma(h_1, a_1) = \sigma(h_2, a_2)$ then $h_1 = h_2$ and $a_1 = a_2$; and
- $u = (u_1, \ldots, u_n)$, where $u_i : Z \to \mathbb{R}$ is a real-valued utility function for player i on the terminal nodes Z.

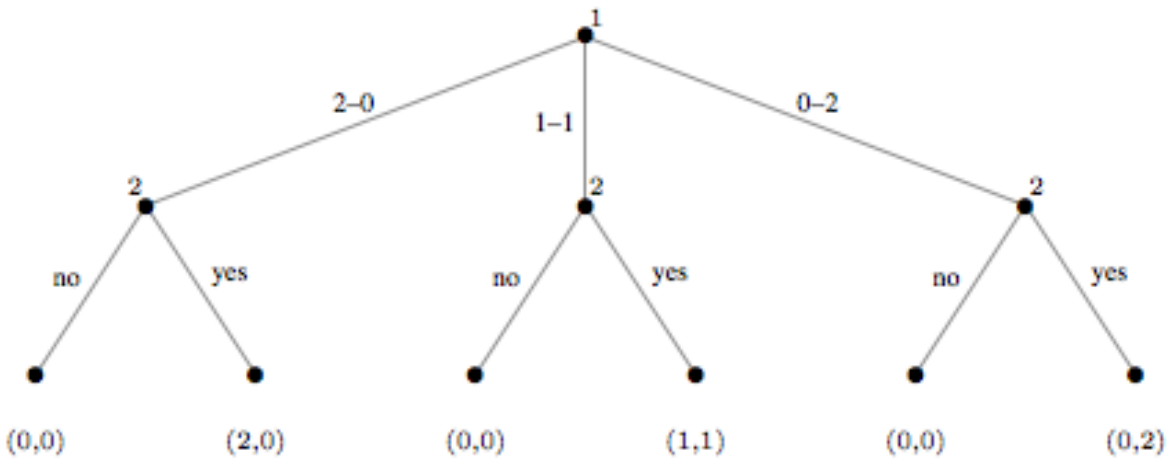
Extensive form games with perfect information

- Since the choice nodes form a tree, we can unambiguously identify a node with its *history*—the sequence of choices leading from the root node to it.
- We can also define the *descendants* of a node h—all the choice and terminal nodes in the subtree rooted at h.

Example: Sharing game

- A brother and sister have to share two indivisible and identical presents.
- Both siblings value the two presents equally and additively.
- They use the following protocol:
  - First the brother suggests a split, which can be one of three—he keeps both, she keeps both, or they each keep one.
  - Then the sister chooses whether to accept or reject the split.
  - If she accepts they each get their allocated present(s), and otherwise neither gets any gift.

Example: Sharing game



Strategies

- A pure strategy for a player in a perfect-information game is a complete specification of which deterministic action to take at every node belonging to that player.

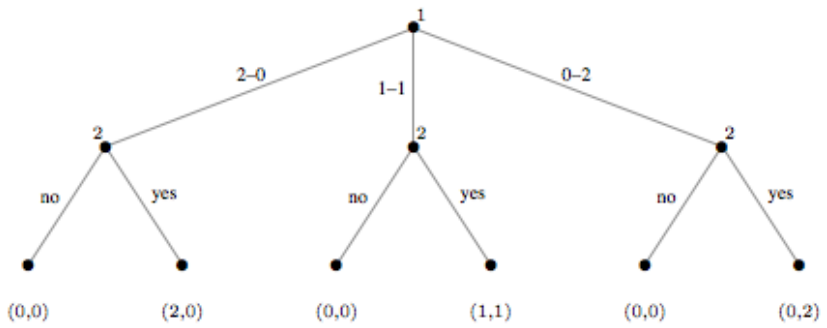
Definition

Let $(N, A, H, Z, \chi, \rho, \sigma, u)$ be a perfect-information extensive form game. Then the set of pure strategies of player i is given by

$$\prod_{h \in H, \rho(h)=i} \chi(h)$$

- Notice that an agent's strategy requires a decision at each choice node, regardless of whether or not it is possible to reach that node given the other choice nodes.

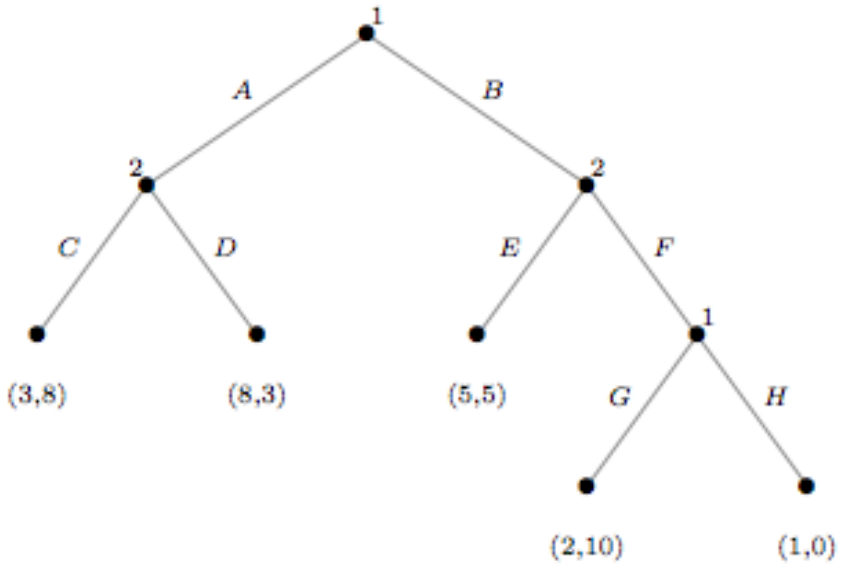
Strategies: Sharing game



- Player 1 has 3 strategies, and player 2 has 8:

$$
\begin{aligned}
S_1 &= \{2-0, 1-1, 0-2\} \\
S_2 &= \{(\text{yes}, \text{yes}, \text{yes}), (\text{yes}, \text{yes}, \text{no}), (\text{yes}, \text{no}, \text{yes}), (\text{yes}, \text{no}, \text{no}), \\
&\quad (\text{no}, \text{yes}, \text{yes}), (\text{no}, \text{yes}, \text{no}), (\text{no}, \text{no}, \text{yes}), (\text{no}, \text{no}, \text{no})\}
\end{aligned}
$$

One more example



$$
\begin{aligned}
S_1 &= \{(A,G),(A,H),(B,G),(B,H)\} \\
S_2 &= \{(C,E),(C,F),(D,E),(D,F)\}
\end{aligned}
$$

One more example: induced normal form

	(C, E)	(C, F)	(D, E)	(D, F)
(A, G)	3, 8	3, 8	8, 3	8, 3
(A, H)	3, 8	3, 8	8, 3	8, 3
(B, G)	5, 5	2, 10	5, 5	2, 10
(B, H)	5, 5	1, 0	5, 5	1, 0

Extensive vs. normal form

- For every perfect-information game there exists a corresponding normal form game.
- However,the temporal structure of the extensive form representation can result in a certain redundancy within the normal form.
  - In the previous example, the normal form has 16 different outcomes, while there are only 5 in the extensive form.
- On the other hand, the reverse transformation—from the normal form to the perfect-information extensive form—does not always exist.
  - Try with Prisoner's Dilemma.

Why only pure strategies?

Theorem

Every (finite) perfect-information game in extensive form has a pure strategy Nash equilibrium.

Proof idea:

- Since players take turns, and everyone gets to see everything that happened thus far before making a move, it is never necessary to introduce randomness into action selection in order to find an equilibrium.
- Can be shown by backward induction.

$\square$

- Both this intuition and the theorem do not hold for more game general classes such as *imperfect-information* games.

Subgame-perfect equilibrium

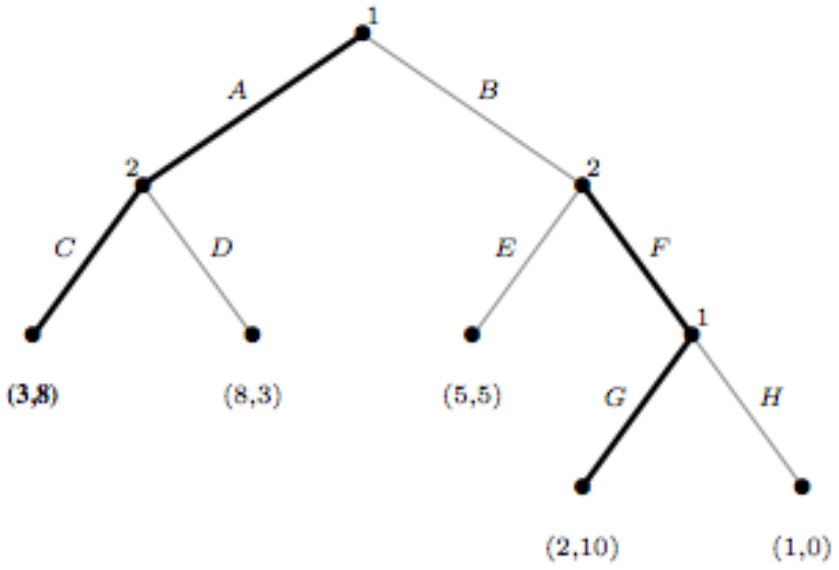
- Refines the concept of Nash equilibrium for extensive form games to reflect the game's temporal structure.

Back to our example

	(C, E)	(C, F)	(D, E)	(D, F)
(A, G)	3, 8	**3**, **8**	8, 3	8, 3
(A, H)	3, 8	**3**, **8**	8, 3	8, 3
(B, G)	5, 5	2, 10	5, 5	2, 10
(B, H)	**5**, **5**	1, 0	5, 5	1, 0

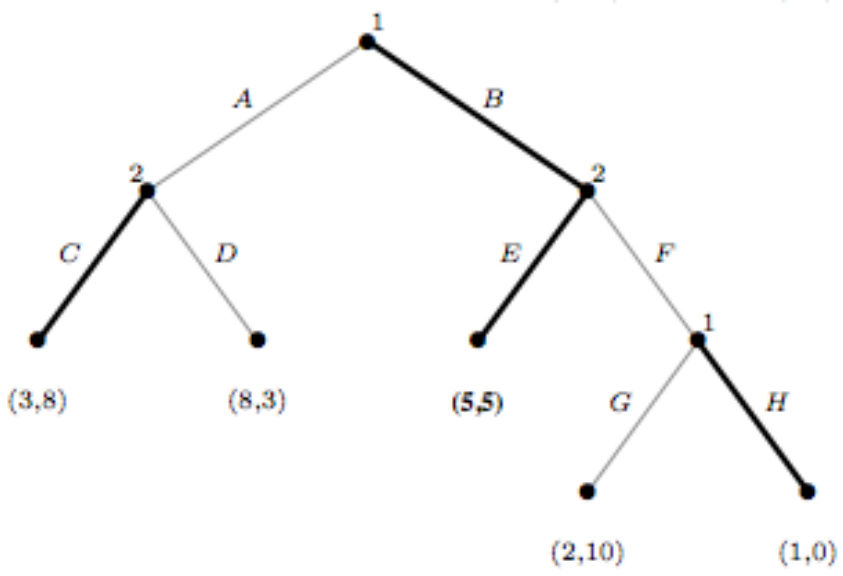
- Three Nash equilibria: $\{(A, G), (C, F)\}$, $\{(A, H), (C, F)\}$ and $\{(B, H), (C, E)\}$.

Back to our example



$$\{(A, G), (C, F)\}$$

Back to our example



$$\{(B, H), (C, E)\}$$

Subgame-perfect equilibrium

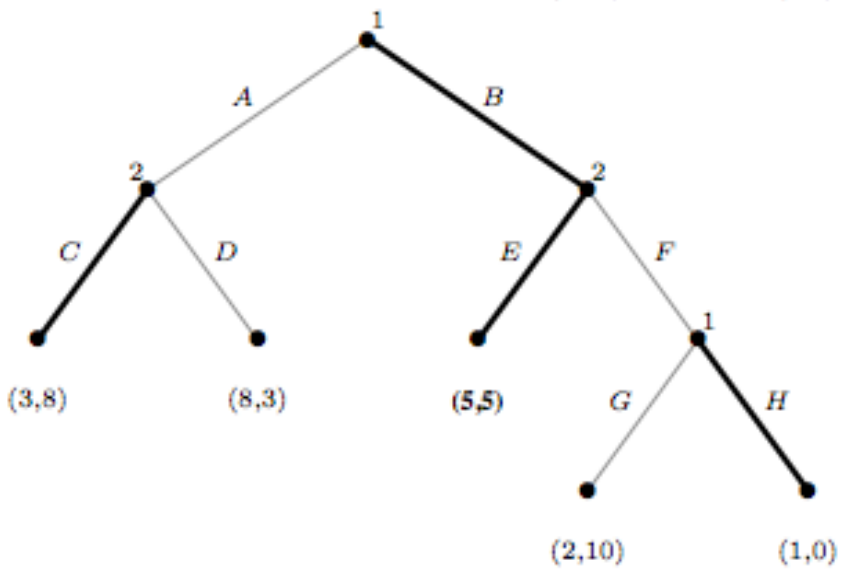
- Given a perfect-information extensive form game G, the **subgame** of G rooted at node h is the restriction of G to the descendants of h.
- The set of subgames of G consists of all of subgames of G rooted at some node in G.

Definition

The **subgame-perfect** equilibria (SPE) of a game G are all strategy profiles s such that for any subgame G' of G, the restriction of s to G' is a Nash equilibrium of G'.

Back to our example

- This rules out all "unsatisfying" equilibria.



$$\{(B, H), (C, E)\}$$

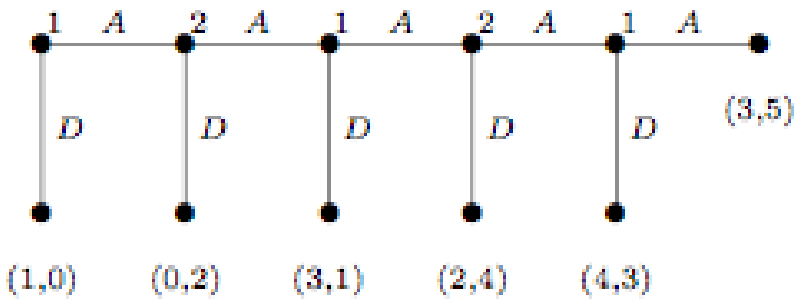
Subgame-perfect equilibrium

- Every SPE is also a Nash equilibrium.
- Even though SPE is a stronger concept than NE, every perfect-information extensive form game has at least one subgame-perfect equilibrium.

Computing SPE: backward induction

- Identify the equilibria in the "bottom-most" subgame trees, and assume that those equilibria will be played as one backs up and considers increasingly larger trees.
- We can use this procedure to compute a sample Nash equilibrium.
  - This way, we are we guaranteed to find a subgame-perfect equilibrium (rather than possibly finding a Nash equilibrium that involves noncredible threats).
  - This procedure is computationally simple—can be implemented as a single depth-first traversal of the game tree and thus requires time linear in the size of the game representation.

Example: Centipede game



- Is the SPE prediction reasonable?

How to sustain cooperation?

	C	D
C	$-1, -1$	$-4, 0$
D	$0, -4$	$-3, -3$

- The unique NE is (D,D). Moreover, this is an equilibrium in dominant strategies!
- In practice, we see many situations of this form, but we often observe some amount of cooperation. Why?

Repeated games

- In many strategic situations, players interact repeatedly.
- Perhaps repetition of the same game might foster cooperation?
- In **repeated games**, the same **stage game** (in normal form) is played multiple times by the same set of players.

Repeated games

- Do agents see what the other agents played earlier?
- Do they remember what they knew?
- While the utility of each stage game is specified, what is the utility of the entire repeated game?

Repeated games

We answer these questions in two steps:

- We first consider the case in which the game is repeated a finite and commonly-known number of times.
- Then we consider the case in which the game is repeated infinitely often, or a finite but unknown number of times.

Finitely repeated games

- One way to represent such games is in extensive form with **imperfect information**.

Extensive form games with imperfect information

- In our discussion of extensive form games we have allowed players to specify the action that they would take at every choice node of the game.
- This implies that players know the node they are in, and—recalling that in such games we equate nodes with the histories that led to them—all the prior choices, including those of other players.
- We might not always want to make such a strong assumption about the players and the environment.
- Imperfect-information games address this limitation.

Extensive form games with imperfect information

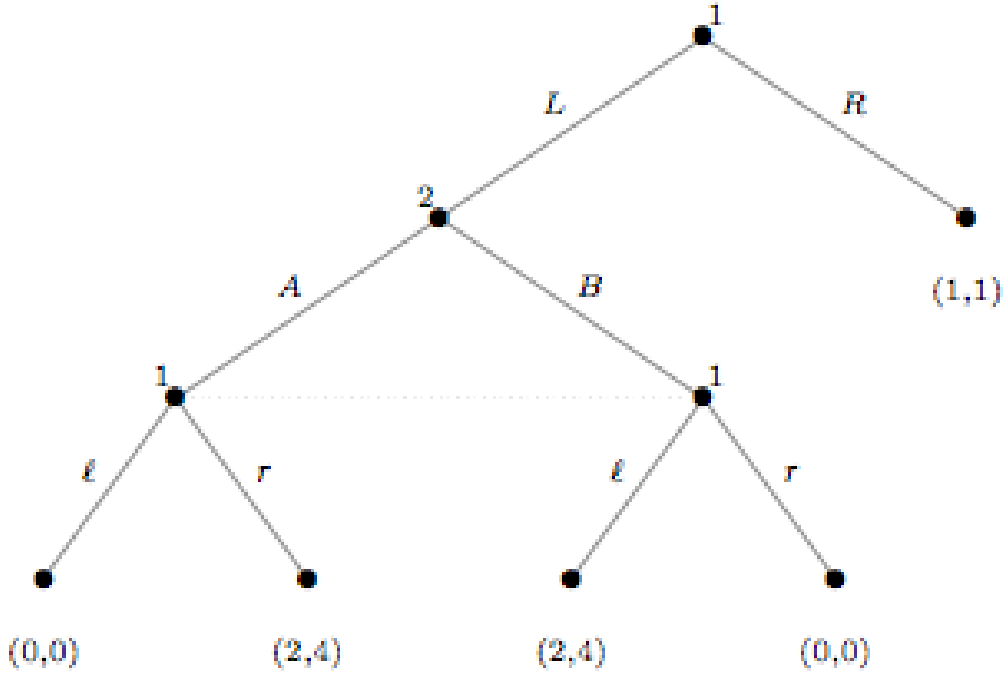
An imperfect-information game is an extensive form game in which each player's choice nodes are partitioned into information sets—if two nodes are in the same information set then the agent cannot distinguish between them.

Definition

A (finite) **imperfect-information game** (in extensive form) is a tuple $(N, A, H, Z, \chi, \rho, \sigma, u, I)$, where:

- $(N, A, H, Z, \chi, \rho, \sigma, u)$ is a perfect-information game;
- $I = (I_1, \ldots, I_n)$, where $I_i = (I_{i,1}, \ldots, I_{i,k_i})$ is a partition of $\{h \in H | \rho(h) = i\}$ such that $\chi(h) = \chi(h')$ and $\rho(h) = \rho(h')$ whenever there exists $j \in \{1, \ldots, k_i\}$ for which $h, h' \in I_{i,j}$.

Example



- Player 1 has 2 information sets: the set including the top choice node and the set including the bottom choice nodes.

Extensive form games with imperfect information

A pure strategy for an agent in an imperfect-information game selects one of the available actions in each information set of that agent.

Definition

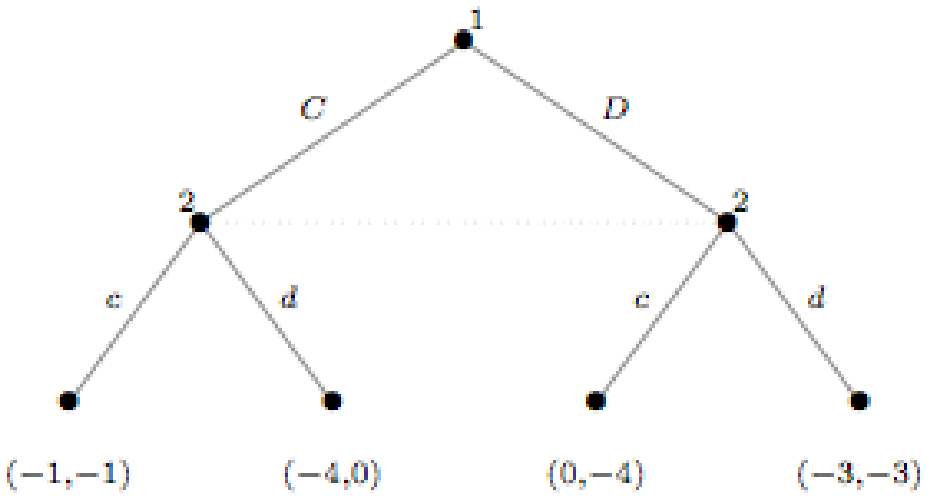
Let $(N, A, H, Z, \chi, \rho, \sigma, u, I)$ be an imperfect-information extensive-form game. Then the set of pure strategies of player i is given by

$$\prod_{I_{i,j} \in I_i} \chi(I_{i,j})$$

where $\chi(I_{i,j}) = \chi(h)$ for any $h \in I_{i,j}$.

- Perfect-information games are a special case of imperfect-information games, in which every element of each partition is a singleton.

Example: Prisoner's Dilemma



- Perfect-information games were not expressive enough to capture the Prisoner's Dilemma game and many other ones.
- In contrast, any normal-form game can be trivially transformed into an equivalent imperfect-information game.
- Imperfect-information games may not always admit a PSNE.

Repeated Prisoner's Dilemma

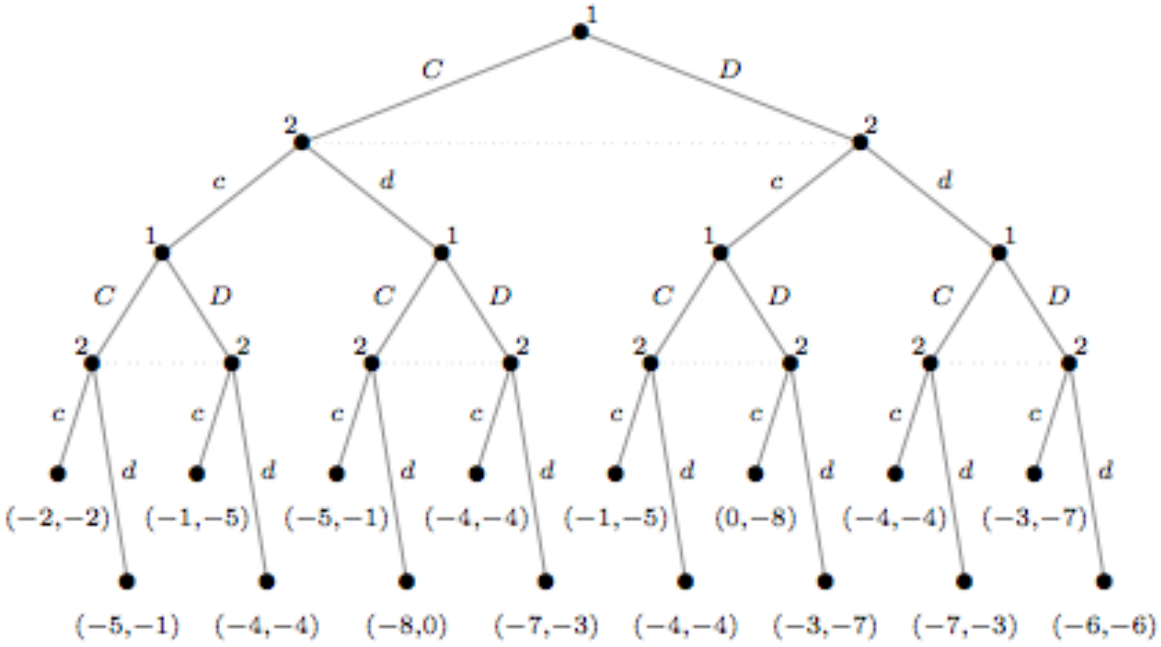
	C	D
C	$-1, -1$	$-4, 0$
D	$0, -4$	$-3, -3$

$\Rightarrow$

	C	D
C	$-1, -1$	$-4, 0$
D	$0, -4$	$-3, -3$

Repeated Prisoner's Dilemma

Twice-played Prisoner's Dilemma in extensive form:



- At each iteration, the players do not know what the other player is playing, but afterwards they do.
- The payoff function of each agent is additive—i.e., the sum of payoffs in the two-stage games.

Strategies in repeated games

- One obvious strategy in the repeated game is to adopt the same strategy in each stage game—this memoryless strategy is called a *stationary strategy*.
- In general, the action (or mixture of actions) played at a stage game can depend on the history of play thus far.
  - This fact plays a particularly important role in infinitely repeated games.
  - In the finite, known repetition case, we encounter again the phenomenon of backward induction (recall the Centipede game). Thus, the only equilibrium in a finitely repeated Prisoner's Dilemma is to always Defect.

Infinitely repeated games

- In extensive form, an infinitely repeated game corresponds to an infinite tree.
  - The payoffs cannot be attached to any terminal nodes.
  - The payoffs cannot be defined as the sum of the payoffs in the stage games (which in general will be infinite).

Infinitely repeated games

There are two common ways of defining a player's payoff in an infinitely repeated game:

- the *average reward* of the stage game in the limit;
- the *future discounted reward*.

Average reward

Definition

Given an infinite sequence of payoffs $u_i^{(1)}, u_i^{(2)}, \ldots$ for player i, the **average reward** of i is

$$\lim_{k \to \infty} \frac{\sum_{j=1}^{k} u_i^{(j)}}{k}$$

Discounted reward

- The future discounted reward to a player at a certain point of the game is the sum of his payoff in the immediate stage game, plus the sum of future rewards discounted by a constant factor.
- This is a recursive definition, since the future rewards give a higher weight to early payoffs than to later ones.

Definition

Given an infinite sequence of payoffs $u_i^{(1)}, u_i^{(2)}, \ldots$ for player i, and a discount factor β with $0 \leq \beta \leq 1$, the **future discounted reward** of i is

$$\sum_{j=1}^{\infty} \beta^j u_i^{(j)}$$

Discounted reward

The discount factor can be interpreted in two ways:

- First, it can be taken to represent the fact that the agent cares more about his well-being in the near term than in the long term.
- Alternatively, it can be assumed that the agent cares about the future just as much as he cares about the present, but with some probability (namely, $1 - \beta$) the game will be stopped any given round.

Strategies in infinitely repeated Prisoner's Dilemma

- **Tit-for-Tat**: a player starts by cooperating (i.e., denying) and thereafter chooses in round $j+1$ the action chosen by the other player in round j.
- **Trigger strategy**: a player starts by cooperating, but if ever the other player defects, then the first defects forever.
- Both Tit-for-Tat and Trigger strategies are in Nash equilibrium for sufficiently large discount factor.
  - Moreover, they form an equilibrium not just with themselves but also with each other.
- This way we can achieve cooperation in each round!

THANK YOU!